

EE 537: Power Quality Problems and Mitigation Techniques in Microgrids

Course Code	EE 537
Course Name	Power Quality Problems and Mitigation Techniques in Microgrids
L-T-P-C	3
Prerequisites	Power System (EE303) & Power Electronics (EE 309)
Intended for	UG/PG
Distribution	3-0-0-3
Mutual Exclusion	NA
Approval	44th BoA (Nov 2021)

Course Contents

- **Power Quality Issues:** Harmonics, frequency deviations, voltage fluctuations, voltage dips, swells, and interruptions. Power tetrahedron, power factor, and other figures of merit under balanced, unbalanced and nonsinusoidal conditions, power quality standards. [8 Lectures]
- **Power Quality Enhancement:** Generation of reference currents/voltages- natural frame of reference (abc), stationary frame of reference ($\alpha\beta 0$) and synchronously reference frame (dq0) and symmetrical components frame reference (+-0). Advanced control architectures for hybrid ACDC Microgrid-decentralized and hierarchical control- droop, primary, secondary and tertiary controls. Cooperative control for power quality enhancement in microgrids-active power Injection, reactive power-sharing, harmonic current sharing and voltage regulation via smart loads. [10 Lectures]
- **Autonomous Control of Distributed Energy Resources in Microgrids:** power sharing schemes for voltage unbalance and harmonics compensation in an Islanded microgrid- Power droop control, virtual impedance loop, local unbalance and harmonics compensation schemes. Effect of faults, overloading and loss of generation on power quality. [8 Lectures]
- **Power Quality Problems associated with Electric Vehicle Charging Infrastructure:** Various configurations of chargers-contact and contact-less chargers, wired AC, DC charging-on-board slow, fast charging, off-board fast, rapid charging. Wireless chargers- inductive, capacitive and hybrid charging topologies. [10 Lectures]
- **Recent Trends and Case Studies:** power quality enhancement in AC-DC hybrid microgrids under grid interactive mode, autonomous control of distributed energy resources, power quality problems and their mitigation techniques for various EV charger topologies. [6 Lectures]

Text Books

1. Arindam Ghosh and Gerard Ledwich, **Power quality enhancement using custom power devices**, Springer Science & Business Media, 2012.
2. Hirofumi Akagi, Edson Hirokazu Watanabe and Mauricio Aredes, **Instantaneous power theory and applications to power conditioning**, John Wiley & Sons, 2017.

References

1. Narain G Hingorani and Laszlo Gyugyi, **Understanding FACTS: concepts and technology of flexible AC transmission systems**, Wiley-IEEE press, 2000.
2. Selected papers, standards and reports